

GiGPO Learn which agent actions matter

Fine-grained credit from trajectories already collected



Episode success hides local mistakes

One signal for the whole episode



A successful detour still gets rewarded.

Compare actions at the same state



Shared search-results page

right item best

wrong item lower

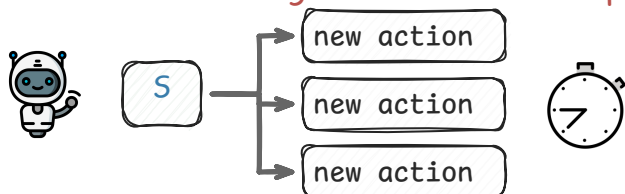
next page worst

Rank observed actions by future return.



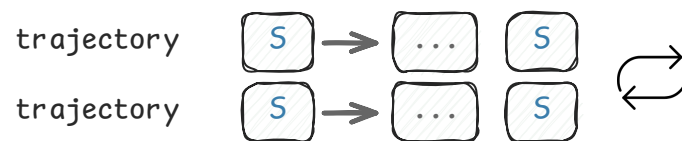
Avoid fresh rollouts at every state

Naive: branch again at each step



Extra generation + environment cost

GiGPO: regroup existing occurrences



Match across episodes AND time steps.

No extra per-state LLM rollouts



Keep the global goal; refine local choices



Episode group

Compare total returns

Task completion signal

+



Anchor-state groups

Compare discounted returns

Action-specific credit



Coherent episodes + better decisions at recurring states



State matching matters: singleton groups add no local contrast.

Critic-free; fine-grained feedback depends on recurring states.